

PyTorch and Graph Neural Networks (GNNs)

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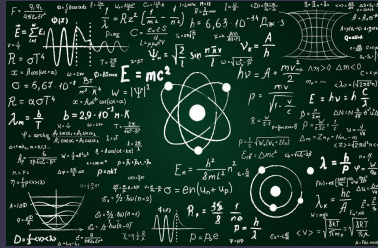
AI IN ASTRONOMY



FAPESP

What does it mean to use ML?

Classically



We get answers
using rules and data!

Machine Learning



We get the rules given
the data and their
answers!

What makes PyTorch so advantageous?

What is PyTorch?

- Machine learning framework produced by Facebook in 2016
- Designed to provide good flexibility and high speeds
- Uses dynamic computation graphs

Advantages

- Pythonic Nature
- Easy to Learn
- Strong Community
- Easy Debugging



 PyTorch  TensorFlow

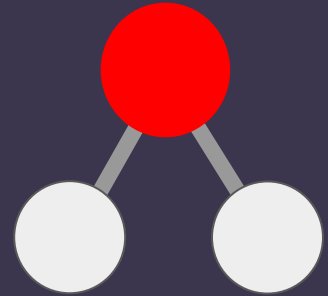
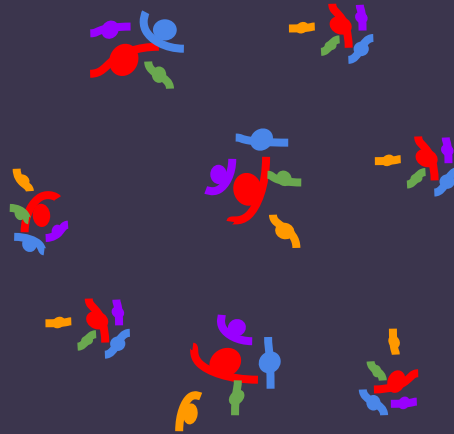
- Dynamic X Static
- **Data parallelism:** asynchronous X manual
- Flexibility, debugging capabilities, and short training duration: >

Different datasets

Tabular data

X	...	Y
15.7	...	24.2
12.1	...	14.6
...	...	...
13.0	...	15.9

"Graph" data



Different datasets

Tabular data



Neural Networks

k Nearest Neighbors

Extremely Randomized Trees

Light Gradient Boosting Machines

Random Forests

Support Vector Machines

"Graph" data



Graph

Neural

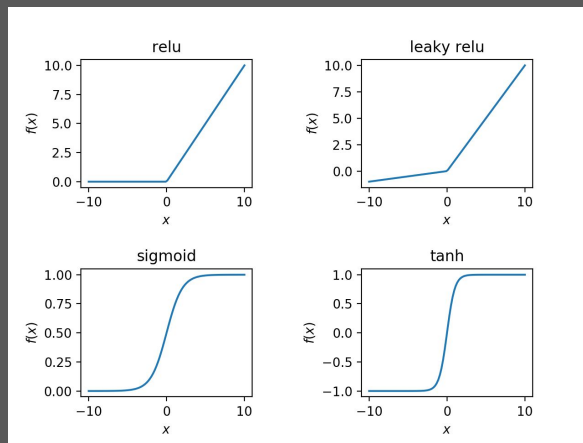
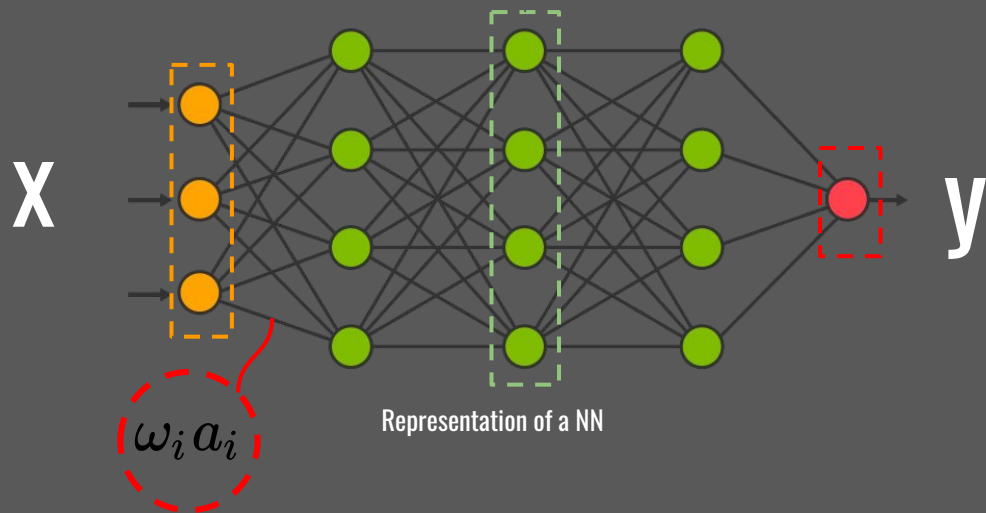
Networks

An overview of Neural Networks

$$b_{\mu} + \sum_{\nu} W_{\mu\nu} a_{\nu}$$

Every layer is responsible to employ an **activation function**:

$$y_{\mu} = f(b_{\mu} + \sum_{\nu} W_{\mu\nu} a_{\nu})$$

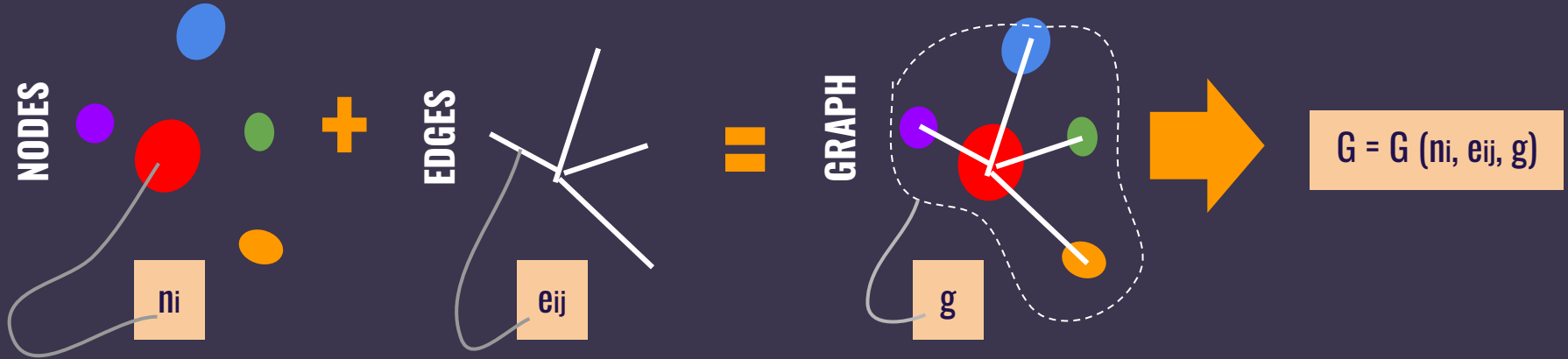


Examples of activation functions

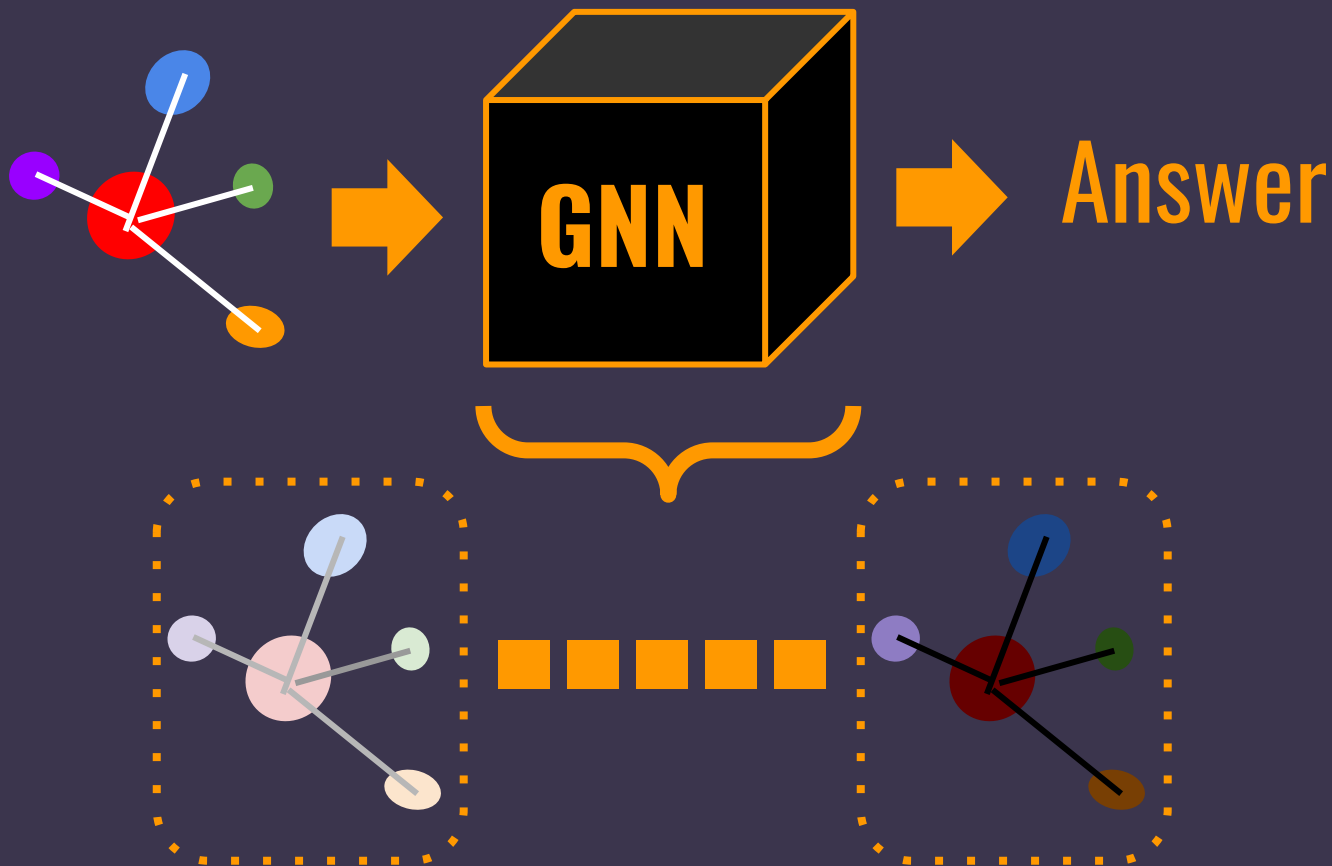
The training process is done is epochs, minimizing the **loss function**, e.g., **Mean Square Error (MSE)**:

$$\text{MSE} = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

What are graphs?

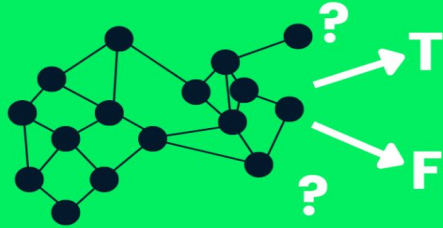


What are graph neural networks?

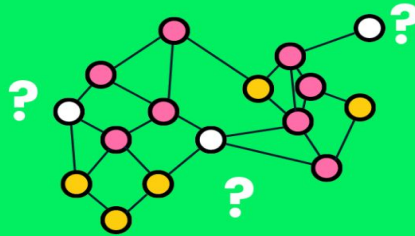


How this answer can be?

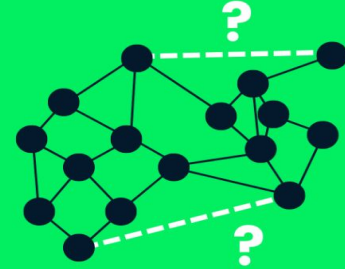
Graph Classification



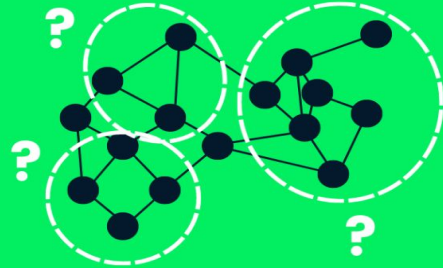
Node Classification



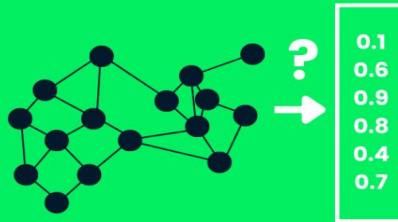
Link Prediction



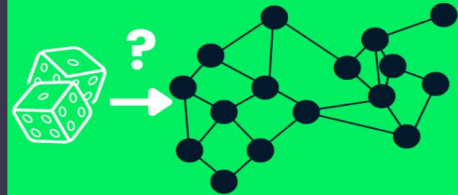
Community Detection



Graph Embedding



Graph Generation



Hands-on activity

STEP 1: go to github and download the repo using:

```
$ git clone https://github.com/natalidesanti/pytorch and GNNs
```

STEP 2: open the **pytorch** notebook in your preferred machine (your own or in Google Colab)

